G H PATEL COLLEGE OF ENGINEERING & TECHNOLOGY

(THE CHARUTAR VIDYA MANDAL (CVM) UNIVERSITY)

DEPARTMENT OF COMPUTER ENGINEERING

HEALTHCARE KNOWLEDGE ASSISTANT USING RAG AND GRAPHRAG

INDUSTRIAL INTERNSHIP (202000801)

MID SEM PRESENTATION

14TH APRIL 2026

PRESENTED BY

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EXTERNAL GUIDE

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AZIRO TECHNOLOGIES PVT. LTD.

PUNE

INTERNAL GUIDE

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Outline

- Company Profile
- Introduction/Problem Statement
- Project Objectives
- Tools and Technologies
- Roles and responsibities
- Work Plan
- Conclusion
- References

Company Profile

- **Company Name:** Aziro Technologies Pvt. Ltd.
- **Location:** Pune, Maharashtra
- **Internship Duration:** 5th January - 30th April

Aziro Technologies is a technology-driven software company focused on:

- AI & Data Science solutions
- Enterprise software development
- Research-oriented problem solving

The internship environment encouraged:

- Independent exploration
- Hands-on learning
- Research-based implementation

INTRODUCTION/MOTIVATION

Patients frequently struggle to:

- Understand medical reports
- Interpret clinical terms
- Find reliable information about diseases

Current healthcare information systems have several limitations:

- Medical knowledge is scattered across research papers
- Existing chatbots often generate inaccurate responses.
- Traditional search systems lack contextual understanding
- Patients struggle to interpret clinical terminology

This project aims to build an AI-powered assistant that can answer healthcare queries using trusted medical knowledge sources.

Project Objectives

The primary objectives of the project are:

- Build a Healthcare Question Answering System.
- Implement Retrieval Augmented Generation (RAG).
- Integrate GraphRAG for relationship-based retrieval.
- Improve answer accuracy using medical knowledge sources.
- Enable users to ask questions related to diseases,drugs and treatments.

System Design

1. **Document Collection**
   Medical documents are gathered for processing
2. **Processing**
   Documents are processed and chunked into manageable parts
3. **Embeddings**
   Embeddings are generated from the processed documents
4. **Knowledge Base**
   A knowledge base is created from the embeddings
5. **User Query**
   User query is processed to retrieve information
6. **Information Retrieval**
   Relevant information is retrieved based on the query
7. **Final Response**
   LLM generates the final response for the user

List of Features/modules

Document Ingestion Module

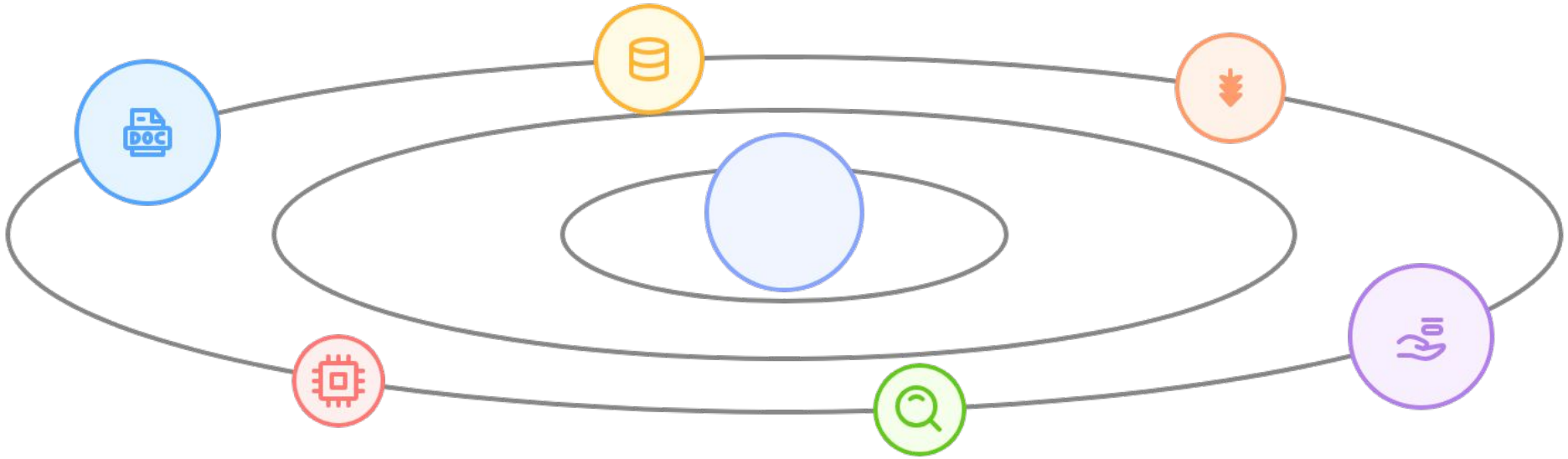
Handles the initial data input into the system

Vector Database Model

Stores data in a vector format for efficient retrieval

LLM Response Generation

Generates responses using large language models



Data Processing Module

Processes and prepares data for further analysis

Retrieval Module

Fetches relevant information from the database

User Interface

The interface for users to interact with the system

Roles/Responsibilities

Internship Overview

Key responsibilities during the internship period

- **Researching** → Researching RAG and GraphRAG architectures
- **Designing** → Designing the knowledge pipeline
- **Collecting** → Collecting healthcare datasets
- **Implementing** → Implementing document ingestion
- **Testing** → Testing retrieval pipelines
- **Evaluating** → Evaluating chatbot responses

Tasks completed so far

RAG Study

Studied RAG and GraphRAG architectures

Datasets Collected

Collected healthcare datasets

Document Extraction

Implemented document extraction pipelines

Ingestion Pipeline

Developed ingestion pipeline for multiple documents

Hybrid Retrieval

Implemented hybrid retrieval (BM25 + embeddings)

Tools & Technologies

Programming Language:

AI / ML Frameworks: LangChain

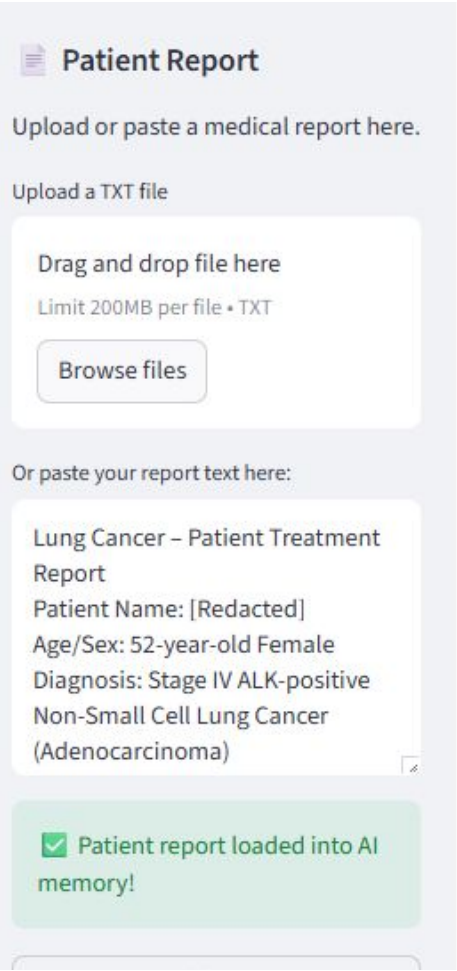
LLMs- Using Groq API

Databases
Vector Database for embeddings (ChromaDB/Qdrant)

Qdrant

Techniques
RAG
GraphRAG(Future Implementation)
Hybrid Retrieval (BM25 + Embeddings)

SCREENSHOTS



Work Plan

Week 1–2
Research RAG & GraphRAG concepts

Week 3–4
Collect datasets and process documents

Week 5
Implement the retrieval pipeline

Week 6
Conduct testing and evaluation

Week 7
Make final improvements to the system

Future Plan

Retrieval Accuracy
Focus on enhancing the accuracy of data retrieval

Healthcare Datasets
Aim to expand the range of healthcare datasets available

GraphRAG
Complete the implementation of GraphRAG framework

Chatbot UI
Develop a user-friendly interface for the chatbot

Performance Evaluation
Conduct evaluations to assess overall system performance

System Documentation
Prepare comprehensive documentation for the final system

References

- Overview of skin cancer types and prevalence rates across continents - ScienceDirect
- CLIP embeddings to improve multimodal RAG with GPT-4 Vision
- LangChain Docs
- OSTEOSARCOMA-TUMOR-ASSESSMENT - The Cancer Imaging Archive (TCIA)
- Welcome - GraphRAG

THANK YOU…